$N=1$ $\qquad Z_0 = \int d\phi \; e^{-\frac{1}{2}a\phi^2}$ $\qquad a>0$

$f: \mathbb{R} \to \mathbb{R}$ $\qquad \phi_1 \equiv \phi$

$\phi \to f(\phi)$

$$\langle f \rangle = \frac{1}{Z_0} \int d\phi \; e^{-\frac{1}{2}a\phi^2} \; f(\phi)$$

$f(\phi) = \phi$

$f(\phi) = \phi^n \qquad n \in \mathbb{N}_0$

$$\langle \phi^n \rangle = \frac{1}{Z_0}$$

$P^2(1,2,3,4)$

$= \{ \underbrace{\{\{1,2\},\{3,4\}\}}, \underbrace{\{\{1,3\},\{2,4\}\}}, \underbrace{\{\{1,4\},\{2,3\}\}} \}$

Z partition function or generating functional

Source

$$Z(g,J) = \int d^N\phi \; e^{-S(\phi) + J^t\phi} \qquad J \in \mathbb{R}^n$$

$$Z(g) = Z(g,0)$$

$$\langle \phi_{i_1} \cdots \phi_{i_n} \rangle = \frac{1}{Z(g)} \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_n}} Z(g,J) \Big|_{J=0}$$

$J^t\phi = \sum_k J_k \phi_k$

For a Gaussian distribution

$$\frac{\partial}{\partial J_j} \frac{\partial}{\partial J_i} e^{J^t\phi} = e^{J^t\phi} \phi_i \phi_j$$

$$Z_0(J) = \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi + J^t\phi}$$

$$\frac{1}{Z_0} \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_n}} Z_0(J) \Big|_{J=0} = \frac{1}{Z_0} \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi} \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_n}} e^{J^t\phi} \Big|_{J=0}$$

$$= \frac{1}{Z_0} \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi} \phi_{i_1} \cdots \phi_{i_n} \underbrace{e^{J^t\phi} \Big|_{J=0}}_{\to 1} = \langle \phi_{i_1} \cdots \phi_{i_n} \rangle$$

$$Z_0(J) = \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi + J^t \phi}$$ complete the square

<u>detour:</u> $N=1$

$$-\frac{1}{2}\phi^2 a + j\phi = -\frac{a}{2}(\phi - A)(\phi - B) + \cdots$$

$$= -\frac{a}{2}\phi^2 - \frac{a}{2}AB + \frac{a}{2}\phi(\underline{A+B}) + \cdots \qquad A = B = j/a$$

$$= -\frac{a}{2}\phi^2 - \frac{1}{2a}j^2 + \phi j + \cdots$$

$$-\frac{1}{2}\phi^2 a + j\phi = -\frac{a}{2}(\phi - j/a)^2 + \frac{1}{2a}j^2$$

For general N: $-\frac{1}{2}\phi^t A\phi + J^t\phi = -\frac{1}{2}(\phi - A^{-1}J)^t A (\phi - A^{-1}J) + \frac{1}{2}J^t A^{-1} J$

$$Z_0(J) = e^{\frac{1}{2}J^t A^{-1} J} \underbrace{\int d^N\phi \; e^{-\frac{1}{2}(\phi - A^{-1}J)^t A (\phi - A^{-1}J)}}$$

translation invariant $\qquad \phi \to \phi + A^{-1}J$

$$= \int d^N\phi \; e^{-\frac{1}{2}\phi^t A \phi}$$

$$= Z_0(0)\, e^{\frac{1}{2}J^t A^{-1} J}$$

$$\langle \phi_{i_1} \cdots \phi_{i_n} \rangle_0 = \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_n}} e^{\frac{1}{2}J^t A^{-1} J} \Big|_{J=0}$$

$$J^t A^{-1} J = \sum_{k\ell} J_k A^{-1}_{k\ell} J_\ell$$

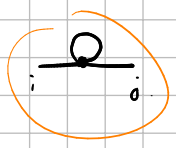
$$\langle \phi_i \phi_j \rangle_0 = \frac{\partial}{\partial J_i}\frac{\partial}{\partial J_j} e^{\frac{1}{2}J^t A^{-1} J} \Big|_{J=0}$$

$$= \frac{\partial}{\partial J_i}\left(e^{\frac{1}{2}J^t A^{-1} J} \, \frac{1}{2}\underbrace{\left(\sum_\ell A^{-1}_{j\ell} J_\ell + \sum_k A^{-1}_{kj} J_k \right)}\right)\Big|_{J=0}$$

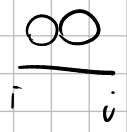
eval at $J=0$

$$= e^{\frac{1}{2}J^t A^{-1} J}\Big|_{J=0} \frac{1}{2}\left(A^{-1}_{ji} + A^{-1}_{ij} \right) + 0 \cdot \frac{\partial}{\partial J_i} e^{\frac{1}{2}J^t A^{-1} J}\Big|_{J=0}$$

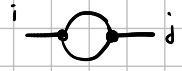
$$= A^{-1}_{ij}$$



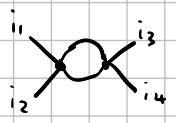
$$= -g \sum_k \phi_i \phi_j \phi_k \phi_k \phi_k \phi_k = -g \sum_k \langle \phi_i \phi_k \rangle_0 \langle \phi_j \phi_k \rangle_0 \langle \phi_k \phi_k \rangle$$



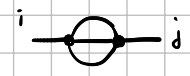
$$= -g \sum_k \phi_i \phi_j \phi_k \phi_k \phi_k \phi_k = -g \sum_k \langle \phi_i \phi_j \rangle_0 \langle \phi_k \phi_k \rangle_0 \langle \phi_k \phi_k \rangle$$

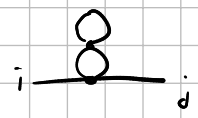


$$= g^2 \sum_{k,l} \phi_i \phi_j \phi_k \phi_k \phi_k \phi_l \phi_l \phi_l \equiv g^2 \sum_{kl} \langle \phi_i \phi_k \rangle_0 \langle \phi_j \phi_l \rangle - -$$

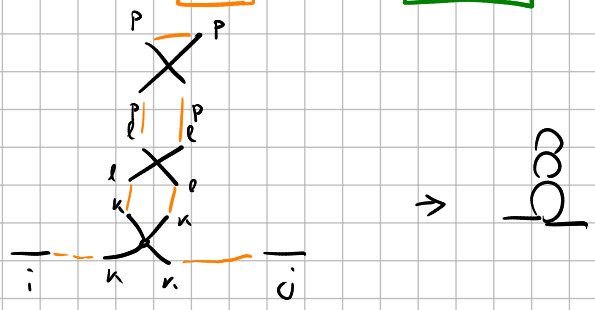
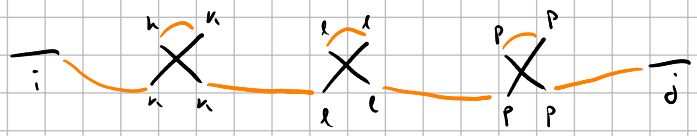
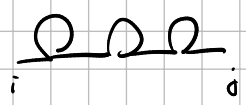


$$= g^2 \sum_{kl} \phi_{i_1} \phi_{i_2} \phi_{i_3} \phi_{i_4} \phi_k \phi_k \phi_k \phi_k \phi_l \phi_l \phi_l \phi_l$$





$$-g^3 \sum_{klp} \phi_i \phi_j \phi_k \phi_k \phi_k \phi_k \phi_l \phi_l \phi_l \phi_l \phi_p \phi_p \phi_p \phi_p$$



$$-g^3 \sum_{klp} \phi_i \phi_j \phi_k \phi_k \phi_k \phi_k \phi_l \phi_l \phi_l \phi_l \phi_p \phi_p \phi_p \phi_p$$